

**[SINGLE CORRECT CHOICE TYPE]**

**Q.1 to Q.10** has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Q.1 If  $f(x) = \begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta) \end{vmatrix}$  and  $f(0) = -2$  then  $\sum_{r=1}^{30} |f(r)|$  equals

(A) 2 (B) 30 (C) 60 (D) 120

Q.2 Let  $a$ ,  $b$  and  $c$  be positive real numbers, then the system of equations in  $x, y, z$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1, \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \quad -\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

has

(A) no solution

(B) unique solution

(C) infinitely many solution

(D) finitely many solutions

Q.3 Let  $\Delta_1 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \Delta_1 \neq 0$

$$\Delta_2 = \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix} \text{ where } b_{ij} \text{ is cofactor of } a_{ij} \quad \forall i, j = 1, 2, 3$$

and  $\Delta_3 = \begin{vmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{vmatrix}$  where  $c_{ij}$  is cofactor of  $b_{ij} \quad \forall i, j = 1, 2, 3$ .

then which one of the following is always correct.

(A)  $\Delta_1, \Delta_2, \Delta_3$  are in A.P.

(B)  $\Delta_1, \Delta_2, \Delta_3$  are in G.P.

(C)  $\Delta_1^2 = \frac{\Delta_3}{\Delta_2}$

(D)  $\Delta_1 = \frac{\Delta_2}{\sqrt{\Delta_3}}$

Q.4  $P$  is an orthogonal matrix and  $A$  is a periodic matrix with period 4 and  $Q = PAP^T$  then  $X = P^T Q^{2005} P$  will be equal to

(A)  $A$

(B)  $A^2$

(C)  $A^3$

(D)  $A^4$

- Q.5 Let  $M = [a_{uv}]_{n \times n}$  be a square matrix of order  $n$  where  $a_{uv} = \sin(\theta_u - \theta_v) + i \cos(\theta_u - \theta_v)$ , then  $M$  is equal to (where  $\overline{M}$  and  $M^T$  are conjugate of  $M$  and transpose of  $M$  respectively.)  
 (A)  $\overline{M}$  (B)  $-\overline{M}$  (C)  $\overline{M}^T$  (D)  $-\overline{M}^T$
- Q.6 Let  $A_r = \begin{bmatrix} r & 2r-1 \\ 0 & \frac{1}{2^r} \end{bmatrix}$ , then the value of  $\lim_{n \rightarrow \infty} \sum_{r=1}^n \det(A_r)$ , is equal to  
 (A)  $\frac{1}{2}$  (B) 1 (C) 2 (D)  $\frac{3}{2}$
- Q.7 Let matrix  $A = \begin{bmatrix} x & y & -z \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$  where  $x, y, z \in \mathbb{N}$ . If  $\det(\text{adj}(\text{adj}(A))) = 2^8 \cdot 3^4$  then the number of such matrices  $A$ , is  
 [Note : adj.  $A$  denotes adjoint of square matrix  $A$ .]  
 (A) 91 (B) 45 (C) 55 (D) 110
- Q.8 Number of all possible symmetric matrices of order  $3 \times 3$  with each entry 0 or 1 and whose trace equals 1, is  
 (A) 24 (B) 48 (C) 192 (D) 512
- Q.9 If  $a > b > c$  and the system of equations  $ax + by + cz = 0$ ,  $bx + cy + az = 0$  and  $cx + ay + bz = 0$  has non-trivial solution, then both the roots of the quadratic equation  $at^2 + bt + c = 0$  are  
 (A) positive (B) opposite in sign (C) imaginary (D) atleast one positive root
- Q.10 The value of  $\alpha$  for which the matrix  $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$  is inverse of matrix  $B = \begin{bmatrix} -\alpha & 14\alpha & 7\alpha \\ 0 & 1 & 0 \\ \alpha & -4\alpha & -2\alpha \end{bmatrix}$  is  
 (A)  $\frac{1}{2}$  (B)  $\frac{1}{3}$  (C)  $\frac{1}{4}$  (D)  $\frac{1}{5}$

### [PARAGRAPH TYPE]

Q.11 to Q.13 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

#### Paragraph for question nos. 11 to 13

Consider two  $3 \times 3$  matrices  $A$  and  $B$  satisfying  $A = \text{adj. } B - B^T$  and  $B = \text{adj. } A - A^T$  (where  $C^T$  denotes transpose of matrix  $C$ .)

- Q.11 If  $A$  is a non-singular matrix, then  $(\det. A)^2 + (\det. B)^2$  is equal to  
 (A) 16 (B) 48 (C) 64 (D) 128

- Q.12 The value of  $|\text{adj.}(2B^{-1})|$  is equal to  
 (A) 1 (B) 16 (C) 64 (D) 128
- Q.13 If A is a non-singular matrix, then AB is equal to  
 (A) I (B) 2I (C) 3I (D) 4I

**[MULTIPLE CORRECT CHOICE TYPE]**

**Q.14 to Q.18** has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

- Q.14 The system of equation  

$$\begin{aligned} x + 3y + 2z &= 6 \\ x + ay + 2z &= 7 \\ x + 3y + 2z &= b \end{aligned}$$
 has  
 (A) unique solution, if  $a = 2$  and  $b \neq 6$ .  
 (B) infinitely many solution, if  $a = 4$  and  $b = 6$ .  
 (C) no solution, if  $a = 5$  and  $b = 7$ .  
 (D) no solution, if  $a = 3$  and  $b = 5$ .
- Q.15 Let M be a non zero  $3 \times 3$  nilpotent matrix of index 3 and I represents identity matrix of order 3, then  
 (A)  $\det\left(\frac{M^2}{2} + M + I\right) = 0$  (B)  $\det\left(\frac{M^2}{2} + M + I\right) \neq 0$   
 (C)  $\det\left(\frac{M^2}{2} - M + I\right) = 0$  (D)  $\det\left(\frac{M^2}{2} - M + I\right) \neq 0$
- Q.16 Consider three matrices  $A = [a_{ij}]_{2 \times p}$ ,  $B = [b_{ij}]_{3 \times n}$  and  $C = [c_{ij}]_{m \times 2}$  such that  $a_{ij} = \frac{i+j}{2}$  and  $b_{ij} = \frac{i-j}{2}$ , then  
 (A) if  $AB = C$ , then value of  $m + n + p$  is equal to 7.  
 (B) if  $BA = C$ , then value of  $m - n + p$  is equal to 3.  
 (C) if  $AB = C$ , then  $c_{12}$  equals  $\frac{1}{4}$ .  
 (D) if  $BA = C$ , then  $c_{32}$  equals 1.
- Q.17 Let  $A = a_{ij}$  be a matrix of order 3 where  $a_{ij} = \begin{cases} x & \text{if } i = j, x \in \mathbb{R} \\ 1 & \text{if } |i - j| = 1 \\ 0 & \text{otherwise} \end{cases}$   
 then which one of the following **hold(s) good**?  
 (A) For  $x = 2$ , A is a diagonal matrix.  
 (B) A is a symmetric matrix.  
 (C) For  $x = 2$ ,  $\det. A$  has the value equal to 4.  
 (D) Let  $f(x) = \det. A$ , then the function  $f(x)$  has both local maxima and local minima.



Q.18 Let  $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & -1 & 2 \end{bmatrix}$  be a given matrix. If  $P = A^4 + 8A$  and  $Q = -5A^3 + 8A^2 - 10A$ , then

(A)  $\text{tr.}(P + Q) = 10$

(B)  $\text{tr.}(P + Q) = 0$

(C)  $\det.(P + Q) = 8$

(D)  $\det.(P + Q) = 32$

[Note :  $\text{tr.}(M)$  denotes the trace of matrix  $M$  and  $\det.(M)$  denotes determinant of matrix  $M$ .]

### [INTEGER TYPE]

Q.19 to Q.21 are "Integer Type" questions. (The answer to each of the questions are upto 4 digits)

Q.19 Let  $\alpha_1, \alpha_2$  and  $\beta_1, \beta_2$  be the roots of  $ax^2 + bx + c = 0$  and  $px^2 + qx + r = 0$ , respectively. If the system of equations  $\alpha_1 y + \alpha_2 z = 0$  &  $\beta_1 y + \beta_2 z = 0$  has a non-trivial solution, then find the

value of  $\left( \frac{b^2}{q^2} - \frac{ac}{pr} \right)$ .

Q.20 Let  $A = [a_{ij}]_{3 \times 3}$  be a scalar matrix whose elements are the roots of the equation  $x^9 - 6x^8 + 12x^7 - 8x^6 = 0$  and  $B = \text{diag}(b_{11}, b_{22}, b_{33})$  be a diagonal matrix

where  $b_{rr} = \sum_{j=1}^3 \sum_{i=1}^3 a_{ij} \forall r = 1, 2, 3$ . If  $\frac{|\text{adj. } B|}{|\text{adj.}((\text{adj}(\text{adj } A)))|} = \left( \frac{p}{q} \right)^6$

where  $p, q$  are relatively prime number, then find the value of  $(q - 2p)$ .

[Note :  $\text{adj } P$  denotes adjoint matrix of matrix  $P$ .]

Q.21 Let  $A$  be an orthogonal invertible square matrix of order 3 with  $A = \begin{bmatrix} x^2 + 1 & x & 5 \\ \sin x & 3 - 2x^2 & 3 \\ 2 & \ln x & 1 \end{bmatrix}$

and  $f(x) = \text{tr.}(A^{-1})$ . Let  $B$  be an involutory invertible square matrix of order 2 with  $B = \begin{bmatrix} 1-x & 3 \\ 5 & 2x-1 \end{bmatrix}$

and  $g(x) = \text{tr.}(B^{-1})$ . Let  $C = \begin{bmatrix} 1 & x & 0 \\ 0 & 1 & 0 \\ x & x & 1 \end{bmatrix}$  be an idempotent invertible matrix and  $h(x) = \text{tr.}(C^{-1})$ , then

find the number of points of non derivability of  $\max. (f(x), g(x), h(x))$ .

[Note :  $\text{tr.}(P)$  denotes trace of matrix  $P$ .]

### ANSWER KEY

|      |    |      |     |      |    |      |      |      |    |
|------|----|------|-----|------|----|------|------|------|----|
| Q.1  | C  | Q.2  | D   | Q.3  | C  | Q.4  | A    | Q.5  | D  |
| Q.6  | C  | Q.7  | C   | Q.8  | A  | Q.9  | B    | Q.10 | D  |
| Q.11 | D  | Q.12 | A   | Q.13 | D  | Q.14 | BCD  | Q.15 | BD |
| Q.16 | AB | Q.17 | BCD | Q.18 | AD | Q.19 | 0000 | Q.20 | 2  |
| Q.21 | 3  |      |     |      |    |      |      |      |    |